

Package ‘replicateEverything’

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Title Tools to Replicate Empirical Research Papers

Version 0.2.0

Description Provides tools for discovering and running computational replications of empirical research papers. The package connects to a replication registry, retrieves replication materials (data, code, and metadata), and reproduces figures and tables from published studies in a standardized workflow.

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replicateEverything-package

replicateEverything: Reproduce Empirical Research Results

Description

The replicateEverything package provides tools for discovering and executing computational replications of empirical research papers. It connects to a public replication registry containing metadata, replication scripts, and processed datasets required to reproduce figures and tables from published studies.

Workflow

A typical workflow using the package is:

1. Retrieve metadata for a paper using `get_doi_metadata()`.
2. Search the replication registry using `search_papers()`.
3. Create a template for contributing a replication using `create_replication_template()`.
4. Inspect available replications using `list_replications()`.
5. Run a single replication using `run_replication()`.
6. Reproduce all results from a paper using `replicate_paper()`.

Examples

Retrieve metadata for a paper:

```
get_doi_metadata("10.1177/00491241211036161")
```

Search the registry:

```
search_papers("causal")
```

Create a replication template:

```
create_replication_template("10.1177/00491241211036161")
```

List replications:

```
list_replications("10.1177/00491241211036161")
```

Run a single replication:

```
run_replication("10.1177/00491241211036161","fig_1")
```

Replicate an entire paper:

```
replicate_paper("10.1177/00491241211036161")
```

Registry

Replication metadata and materials are stored in the public registry: <https://github.com/replicate-anything/registry>.

Author(s)

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See Also

Useful links:

- <https://github.com/replicate-anything/replicateEverything>
- <https://replicate-anything.github.io/replicateEverything/>
- Report bugs at <https://github.com/replicate-anything/replicateEverything/issues>

artifact_available	<i>Check whether a precomputed artifact is available</i>
--------------------	--

Description

Check whether a precomputed artifact is available

Usage

```
artifact_available(doi, what, repo = NULL)
```

Arguments

doi	Character. DOI of the paper.
what	Character. Replication identifier (e.g., "fig_1").
repo	Optional repository slug.

Value

Logical scalar.

Examples

```
## Not run:  
artifact_available("10.1177/00491241211036161", "fig_1")  
  
## End(Not run)
```

artifact_lookup_candidates	<i>Human-readable list of artifact URLs/paths tried for a replication</i>
----------------------------	---

Description

Human-readable list of artifact URLs/paths tried for a replication

Usage

```
artifact_lookup_candidates(doi, what, repo = NULL, folder = NULL)
```

Arguments

doi	Character. DOI of the paper.
what	Replication identifier.
repo	Optional repository slug.
folder	Optional registry folder name from index.csv.

`call_replication_package`*Call a function from a study replication package*

Description

Call a function from a study replication package

Usage

```
call_replication_package(package, fn, ...)
```

Arguments

<code>package</code>	Package name.
<code>fn</code>	Function name as string.
<code>...</code>	Arguments passed to the function.

`clean_replication_source_lines`*Drop roxygen and export-alias lines from replication source for display*

Description

Drop roxygen and export-alias lines from replication source for display

Usage

```
clean_replication_source_lines(lines)
```

Arguments

<code>lines</code>	Character vector.
--------------------	-------------------

```
create_replication_template
```

Create a replication template

Description

Create a replication template

Usage

```
create_replication_template(doi)
```

Arguments

doi Character. DOI of the paper.

Examples

```
## Not run:
create_replication_template("10.1177/00491241211036161")

## End(Not run)
```

```
default_artifact_path
```

Resolve the expected artifact path for a replication entry

Description

Resolve the expected artifact path for a replication entry

Usage

```
default_artifact_path(rep, what)
```

Arguments

rep A single replication entry from replication.yml.
what Replication identifier.

default_format_name	<i>Default format function name for a replication id</i>
---------------------	--

Description

Default format function name for a replication id

Usage

```
default_format_name(rep_id)
```

Arguments

rep_id	Replication identifier.
--------	-------------------------

download_registry_file	<i>Download a registry or package file to a temp path</i>
------------------------	---

Description

Uses `httr` for HTTP(S) URLs to avoid `download.file()` temp-path failures on some Shiny servers.

Usage

```
download_registry_file(url)
```

Arguments

url	HTTP(S) URL or local path.
-----	----------------------------

Value

Character path to temp file.

enrich_package_replication_meta

Merge replication entries from the study package into registry stub metadata

Description

Merge replication entries from the study package into registry stub metadata

Usage

```
enrich_package_replication_meta(meta, ctx)
```

Arguments

meta	Parsed replication metadata.
ctx	Paper context from <code>paper_context()</code> .

Value

Updated metadata list.

ensure_replication_dependencies

Ensure replication package dependencies are available

Description

Ensure replication package dependencies are available

Usage

```
ensure_replication_dependencies(
  replication_meta,
  paper_meta = NULL,
  install_missing = FALSE
)
```

Arguments

replication_meta	A single replication entry from <code>replication.yml</code> .
paper_meta	Optional paper-level metadata list.
install_missing	Logical. Install missing CRAN packages when TRUE.

Value

Invisibly TRUE.

ensure_replication_package

Load or install a study replication package

Description

Tries, in order: local sibling package (when configured), installed package (upgrade from GitHub when outdated), then fresh GitHub install from package_repo_slug().

Usage

```
ensure_replication_package(package, meta = NULL, ctx = NULL)
```

Arguments

package	R package name.
meta	Parsed replication.yml contents.
ctx	Paper context from paper_context().

fetch_package_replication_yaml

Fetch replication.yml from a study package GitHub repo

Description

Lets the app list tables/figures without installing the package.

Usage

```
fetch_package_replication_yaml(meta, ctx)
```

Arguments

meta	Parsed registry stub or package metadata.
ctx	Paper context from paper_context().

Value

Parsed yaml list or NULL.

`find_replication_entry`*Find a single replication entry by id*

Description

Find a single replication entry by id

Usage

```
find_replication_entry(meta, what)
```

Arguments

<code>meta</code>	Parsed replication metadata.
<code>what</code>	Replication identifier.

`find_repo`*Find the repository for a paper replication*

Description

Looks up the replication registry to determine which repository contains the replication materials for a given DOI.

Usage

```
find_repo(doi)
```

Arguments

<code>doi</code>	Character. DOI of the paper.
------------------	------------------------------

Value

Character string containing the GitHub repository name.

Examples

```
## Not run:  
find_repo("10.1177/00491241211036161")  
  
## End(Not run)
```

format_for_display	<i>Apply an optional format function to an analysis object</i>
--------------------	--

Description

When replication.yml defines format, the analysis object (typically from make_*) is passed to format_* for display. Otherwise the object is returned unchanged.

Usage

```
format_for_display(
  object,
  doi,
  what,
  install_deps = FALSE,
  repo = NULL,
  folder = NULL
)
```

Arguments

object	Analysis output to format.
doi	Character. DOI of the paper.
what	Replication identifier.
install_deps	Logical. Install missing dependencies when TRUE.
repo	Optional repository slug.
folder	Optional registry folder name from index.csv.

Value

Object suitable for display (often an HTML string or ggplot).

Examples

```
## Not run:
result <- render_replication("10.1177/00491241211036161", "fig_1")
format_for_display(replication_object(result), "10.1177/00491241211036161", "fig_1")

## End(Not run)
```

format_function_name	<i>Resolve the format function name for a replication entry</i>
----------------------	---

Description

Resolve the format function name for a replication entry

Usage

```
format_function_name(rep)
```

Arguments

rep	A single replication entry from replication.yml.
-----	--

format_script_path	<i>Resolve optional path to a format script</i>
--------------------	---

Description

Resolve optional path to a format script

Usage

```
format_script_path(rep)
```

Arguments

rep	A single replication entry from replication.yml.
-----	--

format_specified	<i>Check whether a replication entry defines a separate format step</i>
------------------	---

Description

Check whether a replication entry defines a separate format step

Usage

```
format_specified(rep)
```

Arguments

rep	A single replication entry from replication.yml.
-----	--

`get_analysis_function` *Resolve the analysis function from a sourced replication script*

Description

Resolve the analysis function from a sourced replication script

Usage

```
get_analysis_function(env, what, type)
```

Arguments

<code>env</code>	Environment containing sourced definitions.
<code>what</code>	Replication identifier.
<code>type</code>	Replication type (figure or table).

`get_artifact_path` *Get artifact URL or local path for a replication*

Description

Get artifact URL or local path for a replication

Usage

```
get_artifact_path(doi, what, repo = NULL, folder = NULL)
```

Arguments

<code>doi</code>	Character. DOI of the paper.
<code>what</code>	Replication identifier.
<code>repo</code>	Optional repository slug.
<code>folder</code>	Optional registry folder name from <code>index.csv</code> .

Value

Character path or URL, or NULL if no artifact is registered.

Examples

```
## Not run:
get_artifact_path("10.1177/00491241211036161", "fig_1")

## End(Not run)
```

get_code	<i>Retrieve replication code for a paper</i>
----------	--

Description

Returns the analysis script and, when defined, the format script.

Usage

```
get_code(doi, what, repo = NULL, folder = NULL)
```

Arguments

doi	Character. DOI of the paper.
what	Character. Replication identifier (e.g., "fig_1").
repo	Optional repository slug.
folder	Optional registry folder name from index.csv.

Details

For package-backed studies, reads inst/replication_code/*.R from the study package GitHub repo when the package is not installed (same idea as reading code/*.R from the registry repo).

Value

A character vector containing the lines of the replication script(s).

Examples

```
## Not run:
head(get_code("10.1177/00491241211036161", "fig_1"))

## End(Not run)
```

get_code_from_package_repo	<i>Build get_code output for a package-backed study from remote source</i>
----------------------------	--

Description

Build get_code output for a package-backed study from remote source

Usage

```
get_code_from_package_repo(meta, ctx, what, package)
```

Arguments

<code>meta</code>	Parsed replication metadata (with replications list).
<code>ctx</code>	Paper context.
<code>what</code>	Replication id.
<code>package</code>	R package name.

<code>get_doi_metadata</code>	<i>Retrieve metadata for a DOI</i>
-------------------------------	------------------------------------

Description

Fetches bibliographic metadata for a research paper using its DOI. The function queries a DOI resolver and returns basic metadata including the paper title, journal, publication year, and authors.

Usage

```
get_doi_metadata(doi)
```

Arguments

<code>doi</code>	Character. DOI of the paper.
------------------	------------------------------

Details

This function is primarily used by `create_replication_template()` to automatically populate metadata in the `replication.yml` file when creating a new replication.

Value

A list containing:

title Title of the paper

journal Journal name

year Publication year

authors Vector of author names

Examples

```
## Not run:
get_doi_metadata("10.1177/00491241211036161")

## End(Not run)
```

`get_replication_meta_impl`*Fetch replication metadata for a paper*

Description

Fetch replication metadata for a paper

Usage

```
get_replication_meta_impl(doi, repo = NULL, folder = NULL)
```

Arguments

<code>doi</code>	Character. DOI of the paper.
<code>repo</code>	Optional repository slug.
<code>folder</code>	Optional registry folder name from <code>index.csv</code> .

Value

Parsed `replication.yml` contents.

`github_package_outdated`*Whether an installed package lags the GitHub ref*

Description

Whether an installed package lags the GitHub ref

Usage

```
github_package_outdated(package, repo, ref = "main")
```

Arguments

<code>package</code>	Installed package name.
<code>repo</code>	GitHub slug <code>org/repo</code> .
<code>ref</code>	Branch, tag, or commit.

github_remote_sha	<i>Latest commit SHA for a GitHub repo ref</i>
-------------------	--

Description

Uses the GitHub REST API (not remotes) so version checks work on servers where remotes::remote_sha() fails on temp paths.

Usage

```
github_remote_sha(repo, ref = "main")
```

Arguments

- | | |
|------|-------------------------|
| repo | GitHub slug org/repo. |
| ref | Branch, tag, or commit. |

infer_result_format	<i>Infer the format of a replication result object</i>
---------------------	--

Description

Infer the format of a replication result object

Usage

```
infer_result_format(object, type)
```

Arguments

- | | |
|--------|--|
| object | Result returned by a replication script. |
| type | Replication type (figure or table). |

Value

Character string describing the result format.

`installed_package_remote_sha`*SHA recorded for a package installed from GitHub*

Description

SHA recorded for a package installed from GitHub

Usage

```
installed_package_remote_sha(package)
```

Arguments

<code>package</code>	Installed package name.
----------------------	-------------------------

Value

Character SHA or NA_character_.

`install_replication_package_github`*Install or upgrade a study package from GitHub*

Description

Install or upgrade a study package from GitHub

Usage

```
install_replication_package_github(package, repo, ref = "main")
```

Arguments

<code>package</code>	R package name.
<code>repo</code>	GitHub slug.
<code>ref</code>	Git ref.

`is_package_replication`*Whether replication metadata refers to an installed R package*

Description

Whether replication metadata refers to an installed R package

Usage

```
is_package_replication(meta)
```

Arguments

`meta` Parsed replication.yml contents.

Value

Logical.

`list_replications`*List available replications for a paper*

Description

List available replications for a paper

Usage

```
list_replications(doi, repo = NULL, folder = NULL)
```

Arguments

`doi` Character. DOI of the paper.
`repo` Optional repository slug.
`folder` Optional registry folder name from index.csv.

Value

A list of replication entries from replication.yml.

Examples

```
## Not run:  
list_replications("10.1177/00491241211036161")  
  
## End(Not run)
```

load_artifact	<i>Load a precomputed artifact for a replication</i>
---------------	--

Description

Load a precomputed artifact for a replication

Usage

```
load_artifact(doi, what, repo = NULL, folder = NULL)
```

Arguments

doi	Character. DOI of the paper.
what	Character. Replication identifier (e.g., "fig_1").
repo	Optional repository slug.
folder	Optional registry folder name from index.csv.

Value

Artifact contents suitable for display, or NULL.

Examples

```
## Not run:  
load_artifact("10.1177/00491241211036161", "fig_1")  
  
## End(Not run)
```

load_artifact_file_path	<i>Load artifact bytes from a local path or registry URL</i>
-------------------------	--

Description

Load artifact bytes from a local path or registry URL

Usage

```
load_artifact_file_path(path)
```

Arguments

path	Local path or HTTP(S) URL.
------	----------------------------

load_index	<i>Load the replication registry index</i>
------------	--

Description

Load the replication registry index

Usage

```
load_index()
```

Value

A data frame containing replication metadata.

Examples

```
## Not run:
head(load_index())

## End(Not run)
```

load_local_replication_data	<i>Load replication data from local paths (for self-contained scripts)</i>
-----------------------------	--

Description

Used at the bottom of replication scripts so they can be run directly from the paper folder without the package orchestrator.

Usage

```
load_local_replication_data(data_paths, paper_dir = getwd())
```

Arguments

data_paths	Character vector of paths relative to the paper folder.
paper_dir	Paper root directory. Defaults to current working directory.

Value

A data frame, list, or other object.

Examples

```
tmp <- tempfile()
dir.create(tmp)
dir.create(file.path(tmp, "data"))
write.csv(
  data.frame(x = 1:3, y = 4:6),
  file.path(tmp, "data", "example.csv"),
  row.names = FALSE
)
load_local_replication_data("data/example.csv", paper_dir = tmp)
```

load_replication_data *Load replication data files*

Description

Supports CSV, RDS, and Stata DTA files from a local registry checkout or remote raw GitHub URLs.

Usage

```
load_replication_data(data_files, ctx)
```

Arguments

data_files	Character vector of paths relative to the paper folder.
ctx	Paper context from <code>paper_context()</code> .

Value

A data frame, a named list of objects, or NULL.

load_replication_package_path
 Load a package from a local source tree

Description

Load a package from a local source tree

Usage

```
load_replication_package_path(path, package)
```

Arguments

path	Path to package root (contains DESCRIPTION).
package	Expected package name.

local_artifact_path	<i>Get the local artifact file path for a replication, if available</i>
---------------------	---

Description

Get the local artifact file path for a replication, if available

Usage

```
local_artifact_path(doi, what, repo = NULL)
```

Arguments

doi	Character. DOI of the paper.
what	Character. Replication identifier (e.g., "fig_1").
repo	Optional repository slug.

Value

Character path or NULL.

make_function_name	<i>Standard make_* function name for a replication id</i>
--------------------	---

Description

Standard make_* function name for a replication id

Usage

```
make_function_name(what)
```

Arguments

what	Replication identifier.
------	-------------------------

`manifest_artifact_paths`*Artifact paths recorded in artifacts/manifest.json*

Description

Artifact paths recorded in artifacts/manifest.json

Usage

```
manifest_artifact_paths(what, ctx)
```

Arguments

<code>what</code>	Replication id.
<code>ctx</code>	Paper context.

`normalize_doi`*Normalize a DOI*

Description

Cleans and standardizes a DOI string so it can be used consistently across package functions. The function removes common DOI URL prefixes and trims whitespace.

Usage

```
normalize_doi(doi)
```

Arguments

<code>doi</code>	Character. A DOI string or DOI URL.
------------------	-------------------------------------

Value

A cleaned DOI string.

Examples

```
normalize_doi("https://doi.org/10.1177/00491241211036161")
```

normalize_html_table	<i>Decode HTML entities in kableExtra / modelsummary table output</i>
----------------------	---

Description

Decode HTML entities in kableExtra / modelsummary table output

Usage

```
normalize_html_table(html)
```

Arguments

html	Character HTML string.
------	------------------------

Value

Character HTML with common entities decoded for browser display.

Examples

```
normalize_html_table("<table>&nbsp;data</table>")
```

package_folder_candidates	<i>Folder names to check when locating a sibling replication package</i>
---------------------------	--

Description

Folder names to check when locating a sibling replication package

Usage

```
package_folder_candidates(package, meta, ctx)
```

Arguments

package	R package name.
meta	Parsed replication.yml contents.
ctx	Paper context from paper_context().

Value

Character vector of folder names (no duplicates).

`package_replication_code_urls`*Raw GitHub URLs for a study package R source file*

Description

Raw GitHub URLs for a study package R source file

Usage

```
package_replication_code_urls(name, repo, ref = "main")
```

Arguments

<code>name</code>	Base name without .R (e.g. <code>make_figure_1</code>).
<code>repo</code>	GitHub slug.
<code>ref</code>	Git ref.

`package_replication_entries`*Replication entries from a study package when the registry stub omits them*

Description

Replication entries from a study package when the registry stub omits them

Usage

```
package_replication_entries(meta)
```

Arguments

<code>meta</code>	Parsed replication metadata.
-------------------	------------------------------

Value

List of replication/prep entries.

```
package_replication_yaml_urls
```

Raw GitHub URLs for a study package replication.yml

Description

Raw GitHub URLs for a study package replication.yml

Usage

```
package_replication_yaml_urls(repo, ref = "main")
```

Arguments

repo	GitHub slug org/repo.
ref	Branch or tag.

Value

Character vector of raw URLs (tried in order).

```
package_repo_ref
```

Git ref for remotes::install_github()

Description

Git ref for remotes::install_github()

Usage

```
package_repo_ref(meta)
```

Arguments

meta	Parsed replication.yml contents.
------	----------------------------------

Value

Character branch, tag, or commit.

package_repo_slug	<i>Resolve GitHub repo slug for a package-backed replication</i>
-------------------	--

Description

Used when no local sibling package is found. Set `paper.package_repo` or top-level repo in `replication.yml` (GitHub slug, e.g. `replicate-anything/rep_10.1371_journal.pone.0278337`).

Usage

```
package_repo_slug(meta, ctx)
```

Arguments

meta	Parsed replication.yml contents.
ctx	Paper context from <code>paper_context()</code> .

Value

Character repo slug.

paper_context	<i>Build base URLs and paths for a paper in the registry</i>
---------------	--

Description

Build base URLs and paths for a paper in the registry

Usage

```
paper_context(doi, repo = NULL, folder = NULL)
```

Arguments

doi	Character. DOI of the paper.
repo	Optional repository slug. Defaults to <code>find_repo(doi)</code> .
folder	Optional registry folder name from <code>index.csv</code> .

Value

A list with `repo`, `folder`, `base_url`, and optional `local_root`.

raw_to_github_browse	<i>Convert a raw GitHub URL to a browseable /blob/ link</i>
----------------------	---

Description

Convert a raw GitHub URL to a browseable /blob/ link

Usage

```
raw_to_github_browse(url)
```

Arguments

url	Raw raw.githubusercontent.com URL.
-----	------------------------------------

Value

Browse URL or original url.

read_lines_url	<i>Read text lines from a local path or HTTP(S) URL</i>
----------------	---

Description

Read text lines from a local path or HTTP(S) URL

Usage

```
read_lines_url(url)
```

Arguments

url	Character URL or file path.
-----	-----------------------------

Value

Character vector of lines, or empty if not found.

`read_package_repo_source`*Read study package source from GitHub (no install required)*

Description

Read study package source from GitHub (no install required)

Usage

```
read_package_repo_source(name, repo, ref = "main")
```

Arguments

<code>name</code>	Base name without .R (e.g. <code>make_figure_1</code>).
<code>repo</code>	GitHub slug.
<code>ref</code>	Git ref.

Value

Character lines or empty.

`read_yaml_url`*Read yaml from an HTTP(S) URL without writing a temp file*

Description

Avoids `download.file()` temp-path failures on some Shiny servers.

Usage

```
read_yaml_url(url)
```

Arguments

<code>url</code>	Character URL.
------------------	----------------

Value

Parsed yaml list or NULL.

```
registry_artifact_rel_paths
```

Relative artifact paths to try under registry papers/<folder>/

Description

Relative artifact paths to try under registry papers/<folder>/

Usage

```
registry_artifact_rel_paths(what, rep = NULL, ctx = NULL)
```

Arguments

what	Replication id.
rep	Optional replication entry.
ctx	Optional paper context (for manifest.json).

```
render_for_display
```

Render a replication and apply formatting for display

Description

Render a replication and apply formatting for display

Usage

```
render_for_display(doi, what, install_deps = FALSE, repo = NULL, folder = NULL)
```

Arguments

doi	Character. DOI of the paper.
what	Character. Replication identifier (e.g., "fig_1").
install_deps	Logical. Install missing CRAN dependencies when TRUE. Defaults to FALSE.
repo	Optional repository slug.
folder	Optional registry folder name from index.csv.

Examples

```
## Not run:
render_for_display("10.1177/00491241211036161", "fig_1")

## End(Not run)
```

render_replication	<i>Render a single replication</i>
--------------------	------------------------------------

Description

Loads data, sources the replication script, and returns a typed result envelope suitable for Shiny display or artifact generation.

Usage

```
render_replication(doi, what, install_deps = FALSE, repo = NULL, folder = NULL)
```

Arguments

doi	Character. DOI of the paper.
what	Character. Replication identifier (e.g., "fig_1").
install_deps	Logical. Install missing CRAN dependencies when TRUE. Defaults to FALSE.
repo	Optional repository slug.
folder	Optional registry folder name from index.csv.

Value

A list with id, type, object, and format.

Examples

```
## Not run:
render_replication("10.1177/00491241211036161", "fig_1")

## End(Not run)
```

replicate_paper	<i>Replicate all results from a paper</i>
-----------------	---

Description

Runs all registered replications (figures and tables) for a given paper.

Usage

```
replicate_paper(doi, install_deps = FALSE)
```

Arguments

doi	Character. DOI of the paper.
install_deps	Logical. Install missing CRAN dependencies when TRUE. Defaults to FALSE.

Value

A named list of replication result envelopes.

Examples

```
## Not run:
replicate_paper("10.1177/00491241211036161")

## End(Not run)
```

replication_error_message

Format a replication error for user-facing display

Description

Unwraps conditionMessage() and, when present, parent errors and the call that failed.

Usage

```
replication_error_message(x)
```

Arguments

x An error condition or character message.

Value

A single character string suitable for logs or UI.

Examples

```
err <- simpleError("Replication failed", call = quote(run_replication()))
replication_error_message(err)
```

replication_index_diagnostics

Report where the replication index was sought (for debugging Shiny)

Description

When a package-backed study lists no tables/figures, this shows which replication.yml URLs were checked. The study index should come from the package repo (e.g. replicate-anything/rep_10.1371_journal not only the registry stub under papers/<folder>/).

Usage

```
replication_index_diagnostics(doi, repo = NULL, folder = NULL)
```

Arguments

doi	Character DOI.
repo	Optional registry repo slug from index.csv.
folder	Optional registry folder name.

Value

A list with registry_sources, package_sources, etc.

Examples

```
## Not run:  
replication_index_diagnostics("10.1177/00491241211036161")  
  
## End(Not run)
```

replication_object	<i>Extract a plain object from a replication result envelope</i>
--------------------	--

Description

Extract a plain object from a replication result envelope

Usage

```
replication_object(x)
```

Arguments

x	A replication result list or raw object.
---	--

Value

The underlying replication object.

Examples

```
result <- list(id = "fig_1", object = data.frame(x = 1), class = "replication_result")  
replication_object(result)
```

resolve_paper_path	<i>Resolve the registry folder name for a paper</i>
--------------------	---

Description

Uses the registry index folder column when available, otherwise derives a path from the normalized DOI.

Usage

```
resolve_paper_path(doi)
```

Arguments

doi	Character. DOI of the paper.
-----	------------------------------

Value

Character folder name under papers/.

resolve_registry_artifact_path	<i>Resolve a precomputed artifact under the registry paper folder</i>
--------------------------------	---

Description

Package-backed studies may ship display artifacts in registry/papers/.../artifacts/ even when code lives in the study package.

Usage

```
resolve_registry_artifact_path(what, ctx, rep = NULL, doi = NULL)
```

Arguments

what	Replication id.
ctx	Paper context.
rep	Optional replication entry.
doi	Optional DOI for caching.

`resolve_replication_package_path`*Resolve a local path to a study replication package*

Description

Search order:

1. Explicit path in `paper.package_path` (if it exists)
2. `getOption("replicateEverything.replication_packages")` map
3. Sibling folders under `replication_packages_root` or monorepo root

Usage

```
resolve_replication_package_path(package, meta, ctx)
```

Arguments

<code>package</code>	R package name.
<code>meta</code>	Parsed replication.yml contents.
<code>ctx</code>	Paper context from <code>paper_context()</code> .

Value

Normalized path, or NULL.

`retry_with_missing_package`*Retry an expression after installing a missing package*

Description

Retry an expression after installing a missing package

Usage

```
retry_with_missing_package(expr, install_missing = FALSE, max_attempts = 2)
```

Arguments

<code>expr</code>	Expression to evaluate.
<code>install_missing</code>	Logical. Whether package installation is allowed.
<code>max_attempts</code>	Maximum number of attempts.

run_replication	<i>Run a single replication</i>
-----------------	---------------------------------

Description

Executes a specific replication (figure or table) for a paper.

Usage

```
run_replication(doi, what, install_deps = FALSE)
```

Arguments

doi	Character. DOI of the paper.
what	Character. Replication identifier (e.g., "fig_1").
install_deps	Logical. Install missing CRAN dependencies when TRUE. Defaults to FALSE.

Value

The underlying replication object.

Examples

```
## Not run:
run_replication("10.1177/00491241211036161", "fig_1")

## End(Not run)
```

save_artifact	<i>Save a replication result as an artifact file</i>
---------------	--

Description

Save a replication result as an artifact file

Usage

```
save_artifact(
  result,
  output_dir,
  doi = NULL,
  repo = NULL,
  folder = NULL,
  install_deps = FALSE
)
```

Arguments

result	A replication result envelope from <code>render_replication()</code> .
output_dir	Directory in which to write the artifact.
doi	Optional DOI; required to apply a registered <code>format_*</code> step.
repo	Optional repository slug.
folder	Optional registry folder name.
install_deps	Logical; passed to <code>format_for_display()</code> .

Value

Invisibly the output file path.

Examples

```
tmp <- tempfile()
dir.create(tmp)
result <- structure(
  list(
    id = "tab_1",
    type = "table",
    object = data.frame(x = 1:2, y = 3:4),
    format = "data.frame",
    meta = list(id = "tab_1")
  ),
  class = "replication_result"
)
save_artifact(result, tmp)
```

search_papers

Search replicated papers

Description

Searches the replication registry for papers matching a keyword in the title or author fields.

Usage

```
search_papers(query)
```

Arguments

query	Character string used to search paper titles or authors.
-------	--

Value

A filtered data frame of matching papers.

Examples

```
## Not run:
search_papers("causal")

## End(Not run)
```

self_run

Run a replication script footer when executed directly

Description

Scripts normally use an if/else footer: `generate_figure <- make_fig_1(read.csv(...))` when run directly, and `generate_figure <- make_fig_1` when sourced by the package.

Usage

```
self_run(make_fn, data_paths, paper_dir = getwd())
```

Arguments

make_fn	Function that accepts data.
data_paths	Paths relative to the paper folder.
paper_dir	Paper root directory.

Examples

```
## Not run:
# At the bottom of a replication script in code/:
# self_run(make_fig_1, "data/fig_1.csv")

## End(Not run)
```

sibling_monorepo_root *Parent directory of the registry when developing in a monorepo*

Description

Parent directory of the registry when developing in a monorepo

Usage

```
sibling_monorepo_root()
```

Value

Character path or NULL.

`sibling_packages_enabled`*Whether to resolve study packages from sibling monorepo folders*

Description

Off by default on servers; enable via `local.R` or `replicateEverything.use_sibling_packages`.

Usage

```
sibling_packages_enabled()
```

`source_replication_functions`*Load only function definitions from a replication script*

Description

Skips the self-run footer so top-level execution (data load + pipe) does not run when the package sources the file.

Usage

```
source_replication_functions(path, env, install_deps = FALSE)
```

Arguments

<code>path</code>	Path to an R script.
<code>env</code>	Environment in which to define functions.
<code>install_deps</code>	Logical. Passed to dependency retry helper.

`source_replication_scripts`*Source replication and optional format scripts into an environment*

Description

Source replication and optional format scripts into an environment

Usage

```
source_replication_scripts(  
  rep,  
  ctx,  
  env,  
  install_deps = FALSE,  
  include_format = TRUE  
)
```

```
try_render_for_display
```

Run a replication and return a result or error object

Description

Like `render_for_display` but never throws; failures are returned as `simpleError` objects.

Usage

```
try_render_for_display(doi, what, install_deps = FALSE, repo = NULL)
```

Arguments

<code>doi</code>	Character. DOI of the paper.
<code>what</code>	Character. Replication identifier (e.g., "fig_1").
<code>install_deps</code>	Logical. Install missing CRAN dependencies when TRUE. Defaults to FALSE.
<code>repo</code>	Optional repository slug.

Value

A replication result list, a display-ready object, or an error.

Examples

```
## Not run:
try_render_for_display("10.1177/00491241211036161", "fig_1")

## End(Not run)
```

```
validate_artifact
```

Validate that a precomputed artifact exists

Description

Validate that a precomputed artifact exists

Usage

```
validate_artifact(doi, what, repo = NULL)
```

Arguments

<code>doi</code>	Character. DOI of the paper.
<code>what</code>	Character. Replication identifier (e.g., "fig_1").
<code>repo</code>	Optional repository slug.

Value

Invisibly TRUE on success.

Examples

```
## Not run:  
validate_artifact("10.1177/00491241211036161", "fig_1")  
  
## End(Not run)
```

validate_paper_artifacts

Validate all artifacts for a paper

Description

Validate all artifacts for a paper

Usage

```
validate_paper_artifacts(doi, repo = NULL)
```

Arguments

doi	Character. DOI of the paper.
repo	Optional repository slug.

Value

Invisibly TRUE if every replication has an artifact.

Examples

```
## Not run:  
validate_paper_artifacts("10.1177/00491241211036161")  
  
## End(Not run)
```

validate_replication	<i>Validate that a replication can be rendered</i>
----------------------	--

Description

Validate that a replication can be rendered

Usage

```
validate_replication(doi, what, install_deps = FALSE)
```

Arguments

doi	Character. DOI of the paper.
what	Replication identifier.
install_deps	Logical. Passed to render_replication().

Value

Invisibly TRUE on success.

Examples

```
## Not run:
validate_replication("10.1177/00491241211036161", "fig_1")

## End(Not run)
```

yaml_url_status	<i>Probe a yaml URL and return a short status string</i>
-----------------	--

Description

Probe a yaml URL and return a short status string

Usage

```
yaml_url_status(url)
```

Arguments

url	Local path or HTTP(S) URL.
-----	----------------------------

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